

Natural materials

Early humans learned to use the materials they found around them to make tools, clothes, and homes. Many natural materials are still used in the same way, while others are combined to make new ones.

Some natural materials come from plants (for example wood, cotton, and rubber), others from animals (silk and wool), or from Earth's crust (clay and metals). Their natural properties—bendy or rigid, strong or weak, absorbent or waterproof—have been put to good use by humans for millions of years. People have also learned to adjust these properties to suit their needs. Soft plant fibers and animal wool are spun into longer, stronger fibers. Animal skins are treated to make leather to wear. Skins were also used to make parchment to write on; now we use paper made from wood. Metals are mixed to make stronger materials called alloys (see pp.62–63).

Materials from animals

Animals, from insects to mammals, are a rich source of materials. The skin of pigs, goats, and cows can be treated and turned into leather. Caterpillars called silkworms spin themselves cocoons that can be unraveled into fine silk threads. Sheep grow thick, waterproof hair that can be cut off, or shorn, and spun into wool thread used for knitting or woven into fabrics.



Silk

Silkworms and their moth parents have been farmed for more than 5,000 years. A cocoon can produce up to 2,950 ft (900 m) of silk thread that can be made into beautiful fabrics.



The silk used for these bright scarves has been dyed. Natural silk is pale in color, and its tone depends on what the silkworms are eating.



Different breeds of sheep produce different types of wool.

Wool

Sheep have been bred for their wool for more than 6,000 years. An average sheep produces wool for about eight sweaters a year—or 60 pairs of socks. Today, wool is often mixed with acrylic fibers.



Wool yarn

Wool is washed, then spun into long fibers, and dyed.

Materials from plants

Plant materials have played a key role in humanity's success as a species. Wood has provided shelters, tools, and transportation, while cotton and flax (a plant used to make linen) have clothed people for thousands of years. Plant materials can be flexible or rigid, heavy or light, depending on the particular combination of three substances in their cell walls: lignin, cellulose, and hemicellulose.



Latex and rubber

Today, a lot of rubber is synthetic, but natural rubber comes from latex, a fluid that can be tapped from certain types of trees. It contains a polymer that makes it elastic.



Cotton

Fluffy cotton, consisting mainly of cellulose, protects the cotton plant's seeds. It is picked and spun into yarn or thread. The texture of cotton fabrics vary depending on how they are woven.



Wood

Different types of wood have different properties, including color, texture, weight, and hardness, making them suitable for different things. Wood pulp is used to make paper. A lot of wood is harvested from wood plantations.